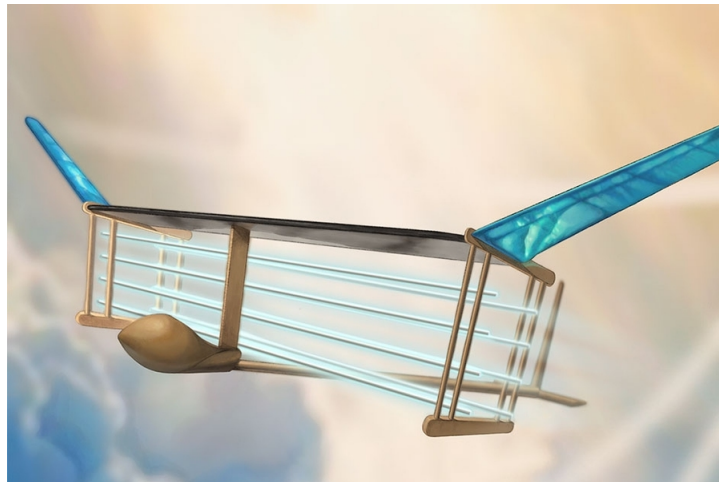


Numerical Simulation Report : Ionic Breeze

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1 Introduction

In a world that needs to be driving towards more sustainability, ecological solutions to fly are certainly future innovations that require our attention. In 2018, the first aircraft to be propelled by ionic wind propulsion performed its first successful flight thanks to MIT engineers. However, as of now, applications are limited due to the difficulties to use such system at a larger scale. Hence, combining this idea with another one could lead to a breakthrough on this subject.

Another system being developed is aerodynamic flow control, a gap in the front or the back of a wing that generates more lift because of the airflow. In this report, we take a look at what can happen if we combine these two concepts, using Fluent numerical simulation. As Ionic Wind propulsion is a very complex phenomenon to model, here we simply simulate the injection of a lighter gas in an aerodynamic flow control gap for a wing.

2 Numerical Simulation

2.1 Geometry and mesh

The wing is modeled as an ellipsis, in which a rectangular gap is created, either at the front or at the back of the wing. The wing is set in a rectangular domain made of air. To be able to inject gas, a rectangular geometry is created underneath the gap, this is the injector.

For the AFC front problem, the ellipsis is 60 meters long and 10 meters large at its center with a gap of 2.5 meters. For the AFC at the back of the wing, the ellipsis is 35 meters long and 6.5 meters large at its center, with a gap of 1.8 meters width.

The other differences between the two setups is that the gap angle is greater for the gap in the front of the wing than for the one in the back and that in the AFC front case, the injector has exactly the same width as the gap while it is slightly smaller in the AFC back case. This causes some of the injected gas to escape behind the injector in the AFC back case.

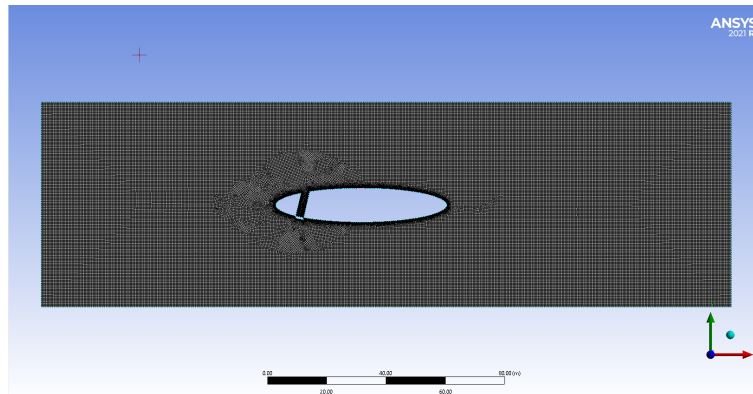


FIGURE 1 – Geometry (AFC front)

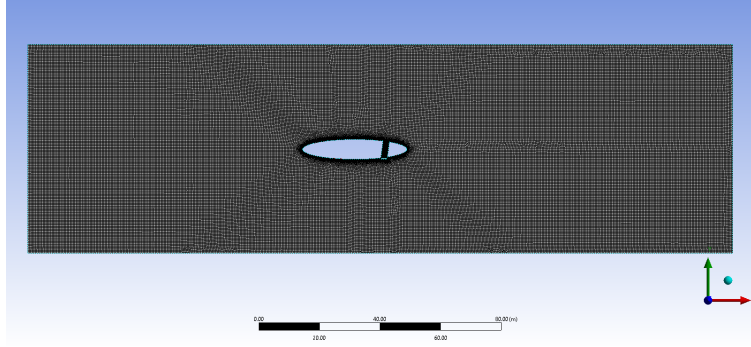


FIGURE 2 – Geometry (AFC back)

To mesh, a quadrangles method is used, with 0.5 m elements in the domain and elements of 0.01m size with an inflation of 5 layers on the edges of the wall and the injector.

2.2 Fluent Setup

Wall boundary conditions are used on the edges of the wing and the injector, except on the injector's edge inside the gap, which is defined as an inlet and that is used to inject the lighter gas. The left boundary of the domain is defined as a velocity inlet, the right one is a pressure outlet while symmetry conditions are applied on the top and bottom.

To simulate this problem, k- ω SST turbulence model is used, as well as species transport. A mixture-template of air and a lighter gas is created. In these simulations, the lighter gas used is hydrogen, which has the following properties : a density of 0.08189 kg/m^3 , a viscosity of $8.411\text{e-}06 \text{ kg/(ms)}$ and a molecular weight of 2.01594 kg/kmol .

The air is supposed to be still, therefore the velocity-inlet on the left boundary of the domain is of 0 m/s and the air fraction on this boundary is 1 (only air). The hydrogen is injected from the top of the injector at 0.1 m/s (the air fraction is set at 0 for this injection- only hydrogen is injected). Lift reports are created at each time step or iteration. A transient formulation is used.

2.3 Results

Here are the results of the simulation. Contours of velocity and density are plotted. It looks like some lift is created. The Y^+ gradient is lesser than 1 which respects the necessary condition to use k- ω SST model. However, there is still some reserve on this model. Most of the residuals are lesser than $1\text{e-}3$, but continuity and energy are not and there are moments in the simulation during which the lift becomes negative.

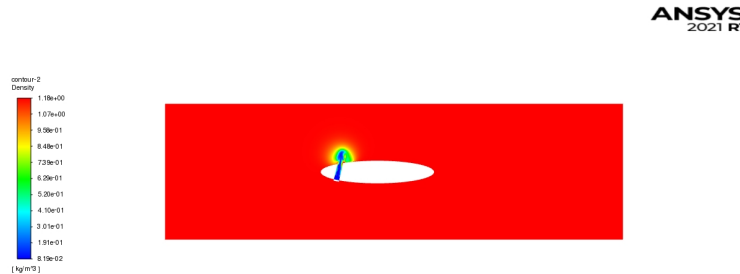


FIGURE 3 – Density (AFC front)

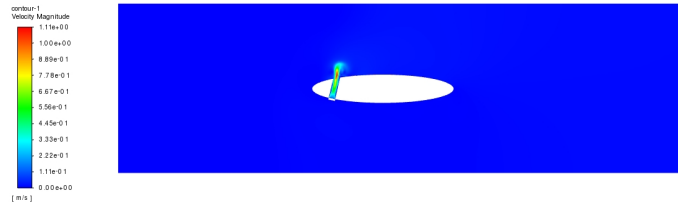


FIGURE 4 – Velocity (AFC front)

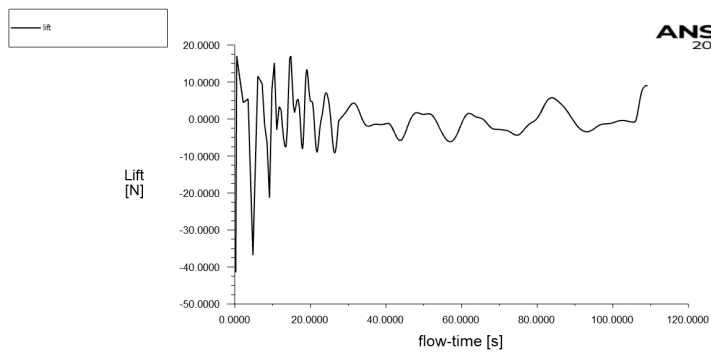


FIGURE 5 – Lift (AFC front)

Console						
(-1.9525012 14.604557 0)		(0.0042559955 0.00089121336 0)			(-1.9465452 14.607448 0)	
Forces - Direction Vector (1 0 0)						
Forces [N]				Coefficients		
Zone	Pressure	Viscous	Total	Pressure	Viscous	Total
well_wing	-1.1960907	0.0038317972	-1.1922589	-1.9525012	0.0042559955	-1.9465452

Net	-1.1960907	0.0038317972	-1.1922589	-1.9525012	0.0042559955	-1.9465452

FIGURE 6 – Thrust (AFC front)

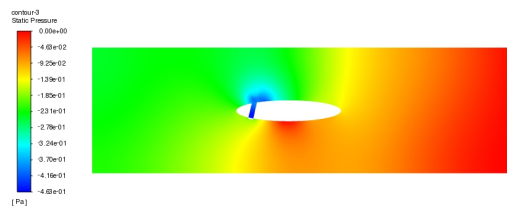


FIGURE 7 – Pressure (AFC front)

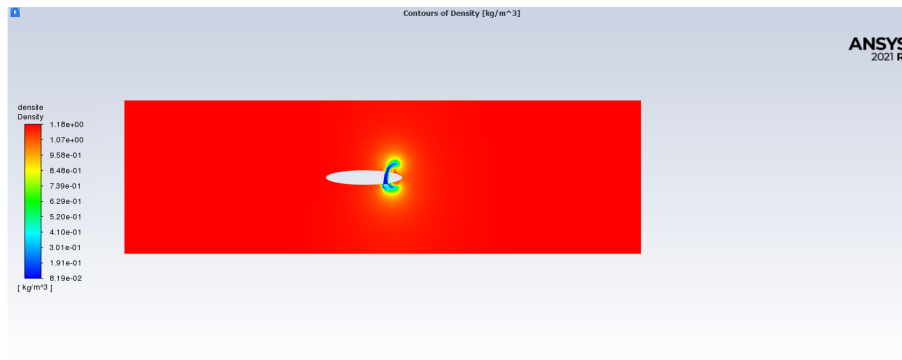


FIGURE 8 – Density (AFC back)

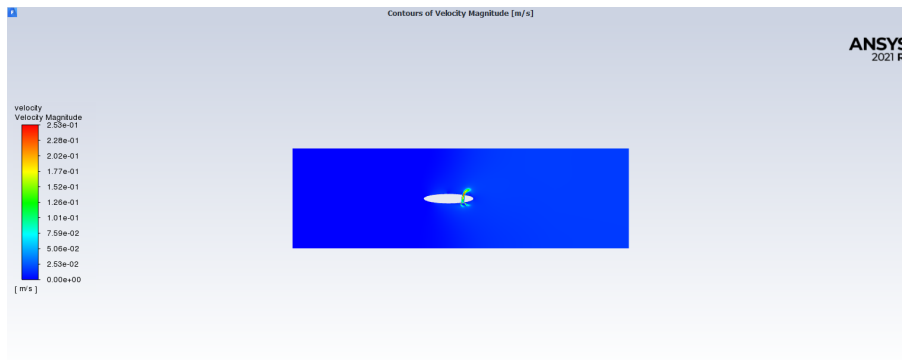


FIGURE 9 – Velocity (AFC back)

0 selected all

Console

(-0.0082763725 0.046270575 0) (-0.00076957418 3.1660736e-06 0) (-0.0090459467 0.046273741 0)

Forces - Direction Vector (1 0 0)

Zone	Forces [N]	Viscous	Total	Coefficients	Viscous	Total
wall_wing	Pressure	-0.0050692782	-0.0055406424	-0.0082763725	-0.00076957418	-0.0090459467
Net		-0.0050692782	-0.0055406424	-0.0082763725	-0.00076957418	-0.0090459467

FIGURE 10 – Thrust (AFC back)

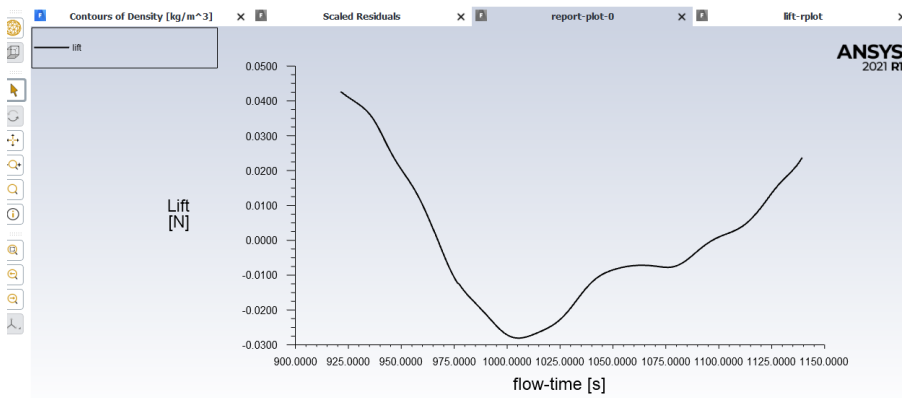


FIGURE 11 – Lift Report (AFC back)

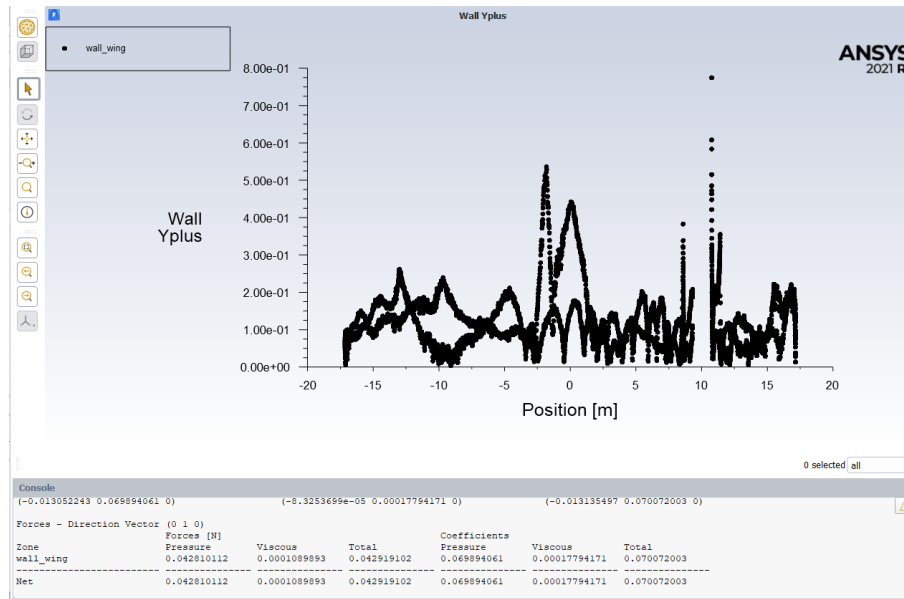


FIGURE 12 – Y^+ (AFC back)

From this figures, it can be assessed that more lift and thrust seem to be created with the AFC in the front and a greater gap angle. It is also possible that having the injector being exactly the same length as the gap influences this result.

3 Conclusion

To recap, this simulation seem to indicate that injecting a lighter gas such as hydrogen in an aerodynamic flow control gap would indeed be a viable solution to generate lift and thrust. However, this is a first rough study using limited computational resources and an oversimplified model. Pursuing simulations efforts with finer model and practical experiments on this subject would be an interesting way to continue research on this subject which has a very high potential value.

4 References

"Study the Influence of a Gap between the Wing and Slotted Flap over the Aerodynamic Characteristics of Ultra-Light Aircraft Wing Airfoil", Cvetelina Velkova¹, Michael Todorov and Guillaume Durand

<https://news.mit.edu/2018/first-ionic-wind-plane-no-moving-parts-1121>.